

FARAFRA, Egypt

WMO: 624230

Lat: 27.0583N

Lon: 27.9892E

Elev: 302

StdP: 14.54

Time Zone: 2.00 (E02)

Period: 03-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.7	40.2	15.7	12.3	69.1	19.2	14.6	65.2	21.2	62.6	18.8	61.8	2.8	210	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.6	106.5	68.4	104.1	68.1	102.2	67.6	72.7	96.6	71.3	96.0	70.2	95.8	9.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.2	83.4	62.7	86.0	83.3	60.6	79.8	83.3	36.4	96.3	35.2	95.8	34.2	96.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.8	15.0	DB	34.1	111.3	1.9	3.7	32.7	114.0	31.5	116.1	30.5	118.2	29.0	120.9	(4)
(5)			WB	28.3	74.7	2.9	1.9	26.3	76.1	24.6	77.2	23.0	78.3	20.9	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.0	54.6	59.0	65.6	73.6	79.9	85.7	87.5	88.0	83.2	76.0	65.3	57.0		
	DBStd	12.72	3.93	6.21	6.37	6.12	5.77	4.10	2.99	2.92	3.96	4.91	4.97	4.04		
	HDD50	8	6	1	0	0	0	0	0	0	0	0	0	1		
	HDD65	881	322	187	65	1	0	0	0	0	0	0	54	252		
	CDD50	8412	150	254	485	708	927	1071	1163	1177	997	805	458	217		
	CDD65	3809	1	20	85	259	462	621	698	712	547	340	62	2		
	CDH74	53060	32	315	1198	3275	6280	9225	10702	10779	7044	3530	635	45		
	CDH80	31267	3	106	495	1658	3639	5855	6816	6831	4043	1641	175	5		
(14) Wind	WSAvg	6.2	5.1	5.5	6.3	6.6	6.9	7.2	6.7	6.2	6.9	6.3	5.3	5.0		
(15) Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecMax	1.1	0.0	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7	0.0		
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	90.7	98.8	102.7	109.6	111.2	108.0	107.3	104.9	100.5	90.4	78.9		
		MCWB	54.5	59.9	62.2	64.0	67.8	70.8	68.9	70.5	67.8	66.3	61.7	58.3		
	2%	DB	72.7	82.7	91.3	98.3	103.4	106.6	104.8	104.8	101.8	95.4	84.6	73.8		
		MCWB	53.6	57.2	59.4	62.4	65.4	67.4	68.5	70.0	67.8	66.1	61.5	55.7		
	5%	DB	69.2	77.0	85.2	94.3	99.5	103.3	102.7	102.7	99.0	91.6	80.8	70.7		
		MCWB	53.3	55.1	57.6	61.0	64.1	67.0	68.0	69.2	67.6	65.9	60.5	55.4		
	10%	DB	66.4	72.8	80.3	89.7	95.4	100.3	100.7	100.6	96.0	88.5	77.1	68.3		
		MCWB	52.3	53.9	56.1	60.0	63.4	66.7	67.9	68.6	67.2	64.8	59.4	54.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	61.5	65.7	67.5	70.0	72.2	73.6	75.5	72.6	71.1	66.4	60.3		
		MCDB	71.0	82.3	90.4	92.8	102.0	101.6	97.1	96.5	91.8	88.0	81.2	73.3		
	2%	WB	55.4	58.7	61.5	64.7	67.2	70.1	71.8	73.2	70.9	69.0	63.8	57.8		
		MCDB	68.9	78.1	84.8	92.4	97.2	99.6	97.0	97.4	92.7	88.3	79.0	70.6		
	5%	WB	54.0	56.6	58.8	62.6	65.4	68.6	70.5	71.8	69.6	67.5	62.1	56.3		
		MCDB	67.4	73.7	81.5	89.3	95.0	98.1	96.7	96.7	92.5	86.7	77.5	68.9		
	10%	WB	52.7	54.8	57.1	60.7	64.1	67.5	69.4	70.4	68.4	66.2	60.5	54.9		
		MCDB	65.9	71.5	78.0	86.9	92.2	96.6	96.1	96.3	91.9	85.7	75.4	67.3		
(35) Mean Daily Temperature Range	MDBR		24.5	25.9	27.1	28.5	28.6	28.4	26.2	25.6	25.1	24.3	23.8	23.8		
	5% DB	MCDBR	28.6	31.8	34.2	35.0	33.5	31.8	28.3	27.5	28.3	27.6	27.6	26.4		
		MCWBR	15.8	15.6	15.4	14.6	13.2	11.2	10.8	10.4	10.7	11.2	13.4	15.0		
	5% WB	MCDBR	25.6	28.9	29.7	30.2	29.6	29.0	25.9	24.7	24.3	24.3	24.0	24.0		
		MCWBR	14.8	15.1	14.7	14.0	12.1	11.1	10.7	10.3	10.4	11.2	12.6	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.393	0.439	0.489	0.576	0.584	0.509	0.487	0.480	0.481	0.461	0.405	0.392		
	taud		2.019	1.877	1.755	1.600	1.606	1.781	1.849	1.874	1.862	1.898	2.017	2.027		
	Ebn at Noon		264	260	255	237	235	252	258	259	254	250	257	257		
	Edh at Noon		47	59	71	85	85	71	66	64	63	57	47	45		
(44) All-Sky Solar Radiation	RadAvg		1346	1635	2004	2264	2458	2582	2531	2372	2121	1764	1440	1257		
	RadStd		33	70	65	53	51	36	21	26	38	22	29	35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page